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PRODUCTION MACHINE LEARNING · CASE STUDY

Moniguard v2: Nigerian PoD Fraud Scoring Engine

BI Scientist • AI/ML Engineer • Python Developer

Calibrated XGBoost + Isolation Forest ensemble scoring pay-on-delivery transactions with PR-AUC 0.877 and 47.9ms p95 latency.

LOGISTICS & FINTECH FRICTION

Why Pay-on-Delivery (PoD) Bleeds Cash in Nigeria

Cash-on-delivery and agent POS transfers represent over 60% of Nigerian e-commerce but suffer from severe fraud vectors.

Delivery-Agent Collusion

- Rider claims customer rejected order, but delivers goods and pockets cash.
- No real-time reconciliation between terminal ping and rider route.
- Losses write off as 'failed deliveries' without audit evidence.

Fake Bank Transfers

- Fraudulent buyers show spoofed SMS / USSD credit alerts.
- Rider releases goods before bank settlement occurs.
- Direct loss borne entirely by the merchant.

Extreme Imbalance

- Fraud represents only ~2.66% of transactions.
- Standard accuracy metric is completely useless (97% accuracy = zero fraud caught).
- Must optimize strictly for PR-AUC and Recall@Top-k%.

ML PIPELINE

74 Features, Chronological Split & Calibrated Ensemble

Zero data leakage with a unified feature pipeline shared across training and FastAPI serving.

Strict Time Split

- 70/30 chronological split on 60,000 transactions.
- Trained on past, tested on future.
- Prevents temporal data leakage.

Hybrid Architecture

- Calibrated XGBoost classifier.
- Isolation Forest for novel zero-day attacks.
- Sigmoid calibration aligns risk probabilities.

Production Serving

- FastAPI `POST /v1/score` endpoint.
- Returns 0–1000 score + top 3 SHAP reasons.
- 47.9 ms p95 response time in Docker.

LIVE OPERATIONS

Moniguard Fraud Workbench

Real-time payment scoring dashboard displaying live risk levels and reason codes.

- **Actionable Reason Codes:** Gives delivery riders and merchant ops immediate explanations (e.g. `HIGH_DISTANCE_DEVIATION`, `NIP_SETTLEMENT_DELAY`).
- **Dynamic Threshold Simulator:** Allows operations managers to tune review thresholds based on seasonal volume spikes.
- **Full MLOps Tooling:** Integrated with MLflow Model Registry, Apache Airflow retraining DAGs, and Docker Compose.

XGBoost FastAPI Docker MLflow



github.com/ikebude/MLOps-Fraud-Detection

CODE-VERIFIED METRICS

Rigorous Evaluation on 60,000 Transactions

All numbers originate from verified `reports/metrics.json` runs with 95% bootstrap confidence intervals.

PR-AUC	ROC-AUC	Recall @ 5% Alert	FastAPI Latency
0.877	**0.943**	**89.2%**	**47.9 ms**
95% CI: 0.845–0.905	Discrimination power	Catches 89% fraud in 5% alerts	p95 benchmark speed

Why PR-AUC Matters

On highly imbalanced datasets (~2.6% fraud), ROC-AUC gives an over-optimistic picture.

PR-AUC measures precision directly against recall across all thresholds.

0.877 PR-AUC ensures operations teams are never flooded with noise.

Production Quality

20/20 Pytest unit and integration tests passing in CI.

Shared preprocessing module eliminates training-serving skew.

Automated Population Stability Index (PSI) drift monitoring.

OPEN-SOURCE & REPRODUCIBLE

Production ML built for real African payment rails.

Complete source code, Docker Compose configuration, feature engineering pipelines, and reproducible benchmark suites available.

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Repository: github.com/ikebude/MLOps-Fraud-Detection